



Name : Yeamin Rahman Mahid
ID: 2023-3-60-375

Submitted to:
Md. Adnan Morshed
Lecturer,
Department of Computer Science
And Engineering

Apply Fourier transform to transform any image from the spatial domain to the

frequency domain. Apply the inverse Fourier transform to transform the image from the

frequency domain to the spatial domain.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
images = [
    "image1.jpg",
    "image2.jpg",
    "image3.jpg"
]
```

```
for file in images:

    # Read image
    img = cv2.imread(file, cv2.IMREAD_GRAYSCALE)

    # Fourier Transform
    f = np.fft.fft2(img)
    f_shift = np.fft.fftshift(f)

    # Magnitude Spectrum
    magnitude = 20 * np.log(np.abs(f_shift) + 1)

    # Inverse Fourier Transform
    img_back = np.fft.ifft2(np.fft.ifftshift(f_shift))
    img_back = np.abs(img_back)

    # Display
```

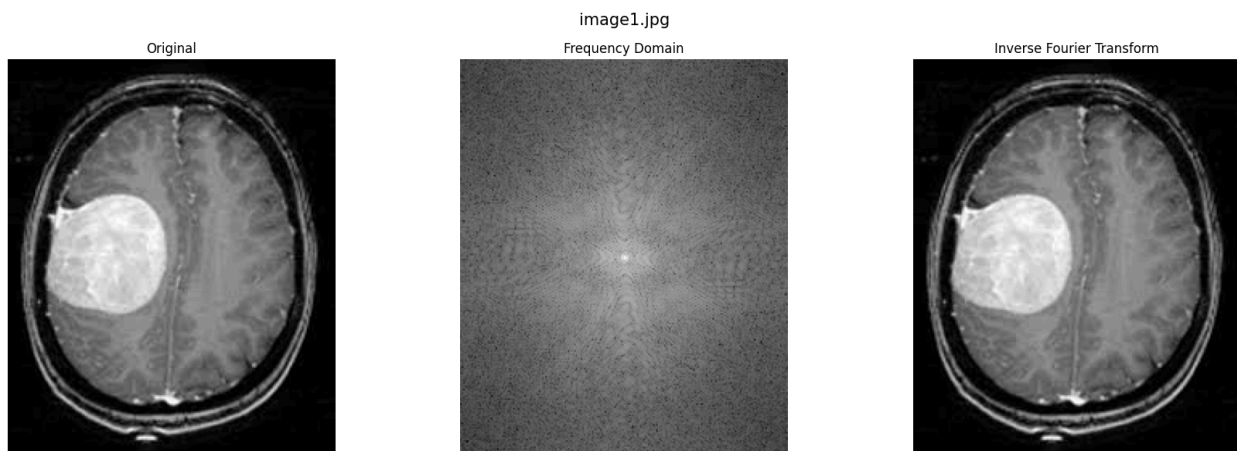
```
plt.figure(figsize=(18,6))

plt.subplot(1,3,1)
plt.imshow(img, cmap='gray')
plt.title("Original")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(magnitude, cmap='gray')
plt.title("Frequency Domain")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(img_back, cmap='gray')
plt.title("Inverse Fourier Transform")
plt.axis("off")

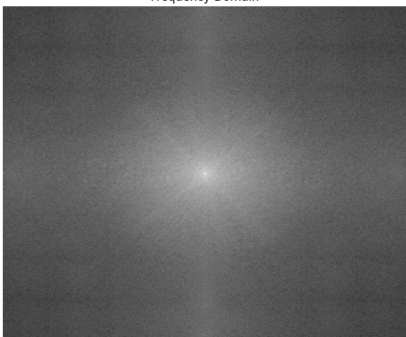
plt.suptitle(file, fontsize=15)
plt.tight_layout()
plt.show()
```



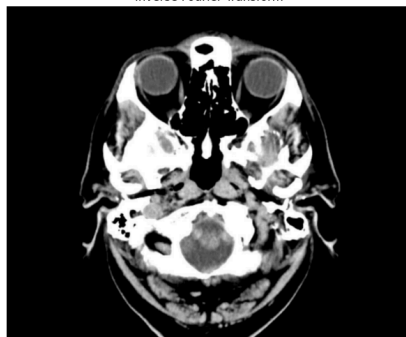
Original



image2.jpg
Frequency Domain



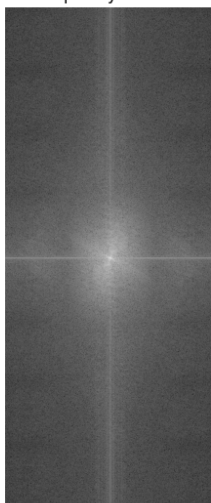
Inverse Fourier Transform



Original



image3.jpg
Frequency Domain



Inverse Fourier Transform

